# Exploring PatchTST: A Transformer Approach to Time Series Forecasting

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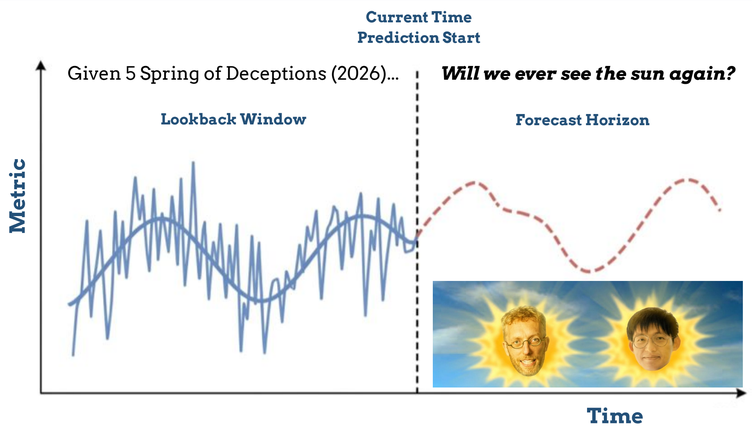
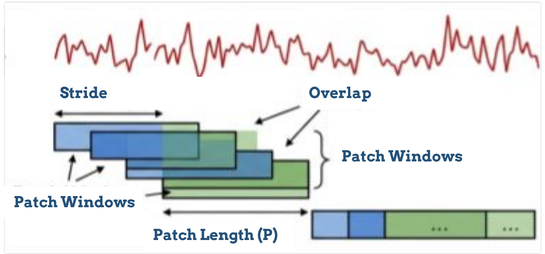
## Introduction



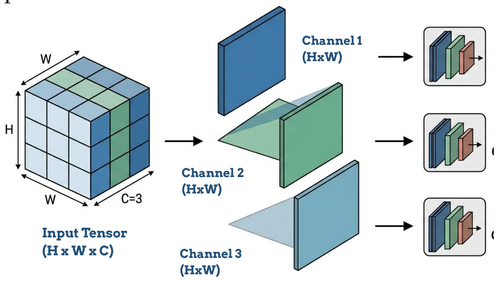

## Background and Motivation

- Transformer models applied to time series tasks were being beat by linear models [2] They are different from traditional NLP tasks
- Our goal is to show that with patching and channel independence, we can show that transformer models outperform linear models on time series tasks

1. Patching

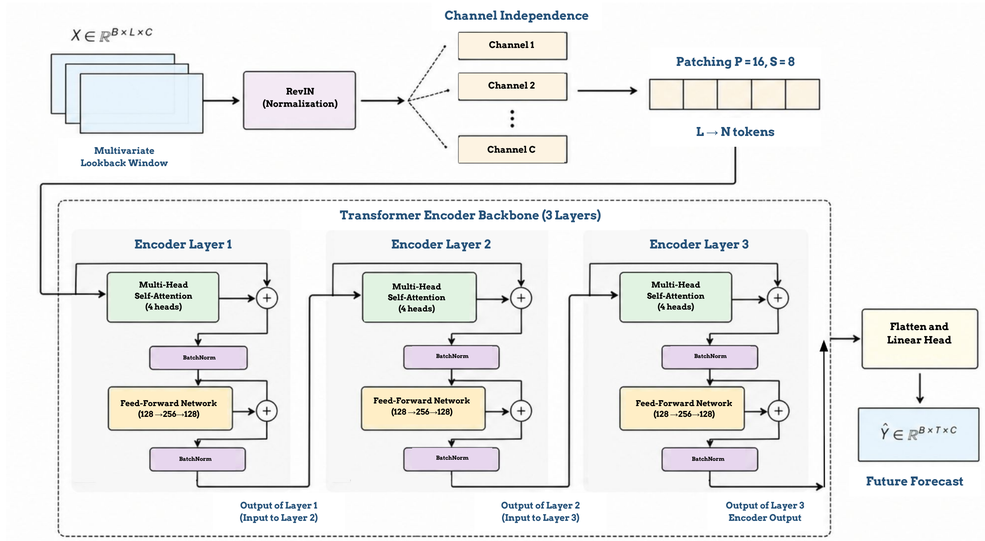



2. Channel Independence

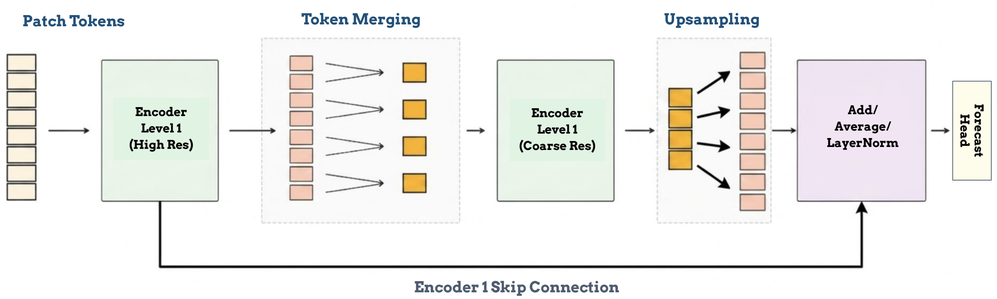



## Methods

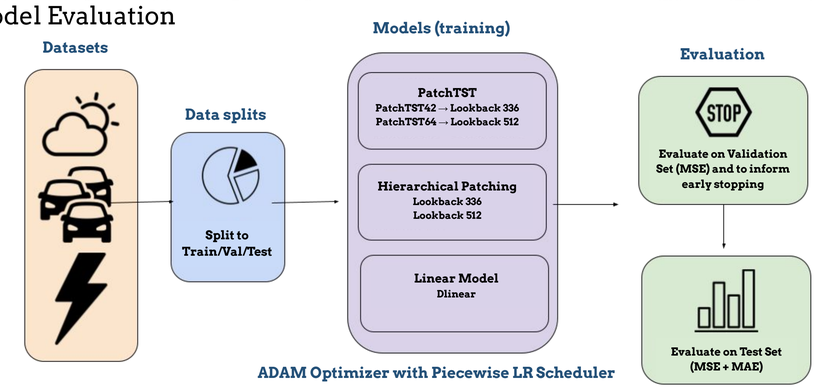
### Architecture




### Hierarchical Patching Independent Exploration (HPatchTST)



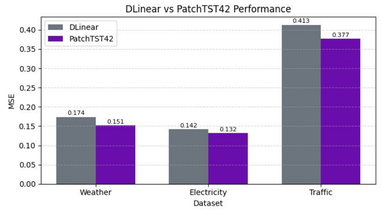

### Model Evaluation



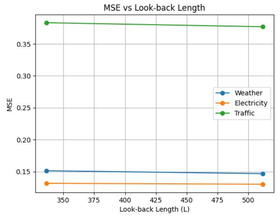

## Results

| | Model | Dataset | | | | | |
|---|---|---|---|---|---|---|---|
| | Pred Len = 96 | Weather | | Electricity | | Traffic | |
| | | MSE | MAE | MSE | MAE | MSE | MAE |
| Ours | **dLinear** | 0.1740 | 0.2349 | 0.1416 | 0.2398 | 0.4129 | 0.2888 |
| | **PatchTST/42 (L=336)** | 0.1513 | 0.1997 | 0.1317 | 0.2258 | 0.3830 | 0.2618 |
| | **PatchTST/64 (L=512)** | **0.1468** | **0.1963** | 0.1302 | 0.2249 | **0.3767** | 0.2602 |
| | **HPatch/42 (L=336)** | 0.1495 | 0.1970 | 0.1319 | 0.2254 | 0.3822 | 0.2643 |
| | **HPatch/64 (L=512)** | 0.1478 | 0.1988 | 0.1301 | 0.2238 | **0.3767** | 0.2627 |
| Paper | **PatchTST/42 (L=336)** | 0.1520 | 0.1990 | 0.1300 | **0.2220** | 0.3670 | 0.2510 |
| | **PatchTST/64 (L=512)** | 0.1490 | 0.1980 | **0.1290** | **0.2220** | 0.3670 | **0.2490** |

PatchTST outperforms Linear Models




Models improve with larger look back (L)

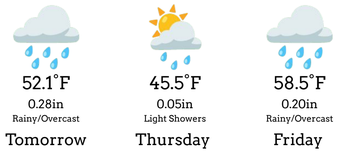



## Conclusion

**Takeaways:**

- PatchTST (and HPatchTST) perform better than linear models
- Reduction in time and space complexity (N = L/S)
- Hierarchical model gains are insignificant from base PatchTST
  - Marginal gains in specific datasets, warrants further study

**So will we see the sun?**




*No, we won't :(*

**Future work:**

- Modelling cross channel dependencies
- Adaptive patching

**References:**

[1] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2022. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. *arXiv:2211.14730*.

[2] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? *arXiv preprint arXiv:2205.13504*, 2022.